Oluwatosin Ayeni

Staff Software Engineer, Inference Infrastructure — Cohere

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PROFILE

Staff-level infrastructure engineer specialising in LLM inference serving at production scale. Designed and operated a full GKE production stack (vLLM + SGLang + LiteLLM Proxy, KEDA autoscaling, 27-target Prometheus, Argo CD GitOps, Terraform IaC) serving 1.25M+ tokens. Benchmarked vLLM and SGLang on 2× NVIDIA H100 NVL 94 GB — 5,965 tok/s peak, 1,443× TTFT speedup via RadixAttention prefix caching, 100% SLO compliance at concurrency 128. Deep expertise in distributed systems, Kubernetes GPU ops, and multi-model routing at scale. Open to London hybrid.

CORE COMPETENCIES

Inference Infrastructure	vLLM, SGLang, TensorRT-LLM, LiteLLM Proxy; production LLM serving at scale — routing, load balancing, fault tolerance, multi-tenant isolation
Kubernetes & GPU Ops	GKE, Helm, Argo CD, Argo Rollouts, KEDA (queue-depth autoscaling), Traefik TLS ingress, Kyverno pod security, NetworkPolicy; NVIDIA L4 and H100 GPU node management
Distributed Systems	Multi-instance request routing, A/B weighted round-robin, Redis response caching, PostgreSQL spend tracking; designed for horizontal scale
Observability & Reliability	27-target Prometheus stack, Grafana 13, AlertManager, Loki, FluentBit DaemonSet, NVIDIA DCGM Exporter — full GPU observability in production
Languages & Tooling	Python, Golang (scripting and tooling), C++ (CUDA kernel profiling), Terraform IaC, Docker, FastAPI, GitHub Actions, SQL

PROFESSIONAL EXPERIENCE

Jan 2025–present — **Founder & Lead Engineer** — *Nexora Academic*, Glasgow, UK

- H100 NVL benchmark: vLLM v0.30 vs SGLang v0.5.20 — Qwen3-8B BF16, TP=2. Peak: 5,965 tok/s; RadixAttention: 1,443× TTFT speedup (39 ms vs 55,565 ms) at 131K-token context; 100% SLO compliance up to concurrency 128. Live: inference-web-theta.vercel.app
- Built *inference-optimizer*: production AI Inference Gateway — 3.87× faster (parallel execution), 61.4% cost reduction (prompt caching), 14× throughput (concurrency), 64.3% cheaper (model routing). 13 optimisations, 4 providers.
- Built *vllm-k8s-stack*: Kubernetes LLM serving — KV-cache-aware routing, HPA autoscaling on GPU pools, Prometheus/Grafana observability.
- Built *inference-harness*: unified benchmark framework spanning closed APIs + vLLM/SGLang/TensorRT-LLM with GPU profiling and live dashboard.
- Designed 13-phase AI Inference Engineering curriculum covering GPU architecture, CUDA optimisation, serving frameworks, and distributed inference.

Jan 2023–present — **Founder & AI Consultant** — *Vertinab / AnaburyAI Technology*, Glasgow, UK

- 28× LLM cost reduction deploying Qwen2.5-7B on RTX 4090 via vLLM/SGLang with continuous batching and KV cache optimisation.
- CUDA KV cache kernel profiling with NVIDIA Nsight: identified memory bandwidth bottleneck; 3.8× throughput improvement; 18× latency reduction via continuous batching redesign.
- Fine-tuned Mistral 7B with LoRA/PEFT on radiology corpus: +22% diagnostic accuracy, −35% hallucination rate; deployed as production clinical inference API.

2024–present — **Senior Software Engineer (Contract)** — *EduTechConnect / Stellar Labs*, Remote

- Real-time NLP pipeline (WhatsApp → Zoho CRM → Supabase) with sub-500 ms end-to-end latency under concurrent load.

Mar 2026–present **AI Automation Engineer** — *EduTech Global*, Remote

- LLM-backed student chatbots with structured output extraction and multi-step reasoning for Ahmadu Bello and Babcock universities.

2018–2022 **Data Analyst** — *Bytelab Technologies Ltd*, Nigeria

KEY PROJECTS

llm-inference-platform (GKE) Production LLM serving: vLLM + SGLang + LiteLLM Proxy on GKE; KEDA autoscaling; 27-target Prometheus; Argo CD GitOps; Terraform IaC. 723 live requests, 1.25M+ tokens. GitHub

llm-inference-benchmark (H100) vLLM vs SGLang, 2× H100 NVLink, Qwen3-8B BF16 TP=2. 5,965 tok/s peak; 1,443× TTFT speedup. Live: inference-web-theta.vercel.app

vllm-k8s-stack Production Kubernetes LLM serving: KV-cache-aware routing, HPA autoscaling, Prometheus/Grafana. GitHub

inference-optimizer Production AI Inference Gateway: 3.87× faster, 61.4% cost reduction, 14× throughput, 64.3% cheaper via model routing. GitHub

inference-harness Unified benchmarking harness for Anthropic, OpenAI, Gemini, vLLM, SGLang, TRT-LLM with GPU profiling. GitHub

EDUCATION

2022–2023 **MSc Artificial Intelligence and Applications, Distinction** — *University of Strathclyde*, Glasgow, UK

2010–2016 **MTech / BSc Physics** — *Federal University of Technology Akure*, Nigeria

OPEN SOURCE

GitHub 43 public repos — github.com/AyeniOluwatosinOlawale